

Stealth-AI: Intelligent Threat Detection, Discreet Dual-Zone Alert, and Anti-Jamming Emergency Response System

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Abstract— Abstract- Stealth-AI is an artificial intelligence (AI) based emergency response and security system for high-priority institutions such as schools and government institutions. It continuously analyzes the real-time CCTV footage using edge-based computer vision and is able to detect threats, suspicious behavior and intrusions with low latency. Upon the detection the system alerts occupants inside and emergency responders outside discreetly to take immediate action without causing too much panic. Unlike conventional systems that are prone to system failure in the event of power outages, network disruptions or jamming attacks, Stealth-AI guarantees continuous operation with edge AI computing, LoRa mesh communication, Ethernet fallback protocols and software defined radio (SDR) based anti-jamming mechanisms. A silent lock-down and visual guidance feature eliminates audible alarms, which may increase chaos by telling occupants to move to safe zones. Additionally, the tracking of intruders by AI is used to produce dynamic heatmaps to aid in accurate situational awareness, coordinated evacuation, and containment. Stealth-AI provides a resilient, autonomous and future proof security solution for critical institutional environments.

Keywords- Edge AI computing, LoRa Mesh Communication, Ethernet Fallback Protocols, Software- defined radio (SDR), Anti-jamming Communication, Decentralized Intelligence, Situational Awareness Systems, AI Heatmap Generation.

I. INTRODUCTION

In Stealth-AI is a hi-tech security and emergency management system designed to operate in high-danger areas such as school buildings and government buildings. It is constantly examining the CCTV footage using AI-based computer vision and uses this to automatically identify threats and send alerts discreetly to personnel and security guards. Its good design enables it to keep functioning even in the absence of power supply when there is a power outage, Internet outage or intentional jamming of the device in the shape of edge AI processing, LoRa mesh network, Ethernet fallback, as well as SDR based anti jamming. Silent and controlled crisis management is more important than detection, which is achieved by Stealth-AI. It may cause silent shutdowns, guide the occupants to places with high risks out of the building by visual indicators, and track the movements of the intruders in real-time with heatmaps. The system is scalable and flexible enough in that detection algorithms, evacuation routes, alert algorithms and the layouts of institutions can be customized depending on the distinctive layouts of the same. It is also dedicated to privacy and compliance, because it is to be ensured that surveillance data remains secured and cannot be

accessed by any internal or unauthorized parties. To be able to implement Stealth-AI successfully, it has to be planned, maintained, and trained accordingly. In order to protect institutions against emerging threats, they should make sure that they have 100 percent CCTVs, periodic diagnostics, special drills, software updates, and firmware updates. The system is more reliable due to strong data protection practice, emergency protocol and communication with the local responders. Through a mix of resilience, silent operation and proactive situation awareness, Stealth-AI will be a reliable future-proof security solution to enhance emergency preparedness in many institutional environments. Its modular design makes it be flexible in the long term to add new sensors, analytics and security technology to fit with the new threat environment. Autonomy + multi-tier communication Stealth-AI develops a new paradigm of smart, resilient institutional safety frameworks.

In Besides the ability to operate, Stealth-AI is concerned with resilience of systems, by decentralizing, intelligence and local decision making. By processing critical the system has edge threat detection and response logic. decreases very massively the field dependency of centralized. servers or permanent internet accessibility. This edge-centric approach guarantees the low latency, speed response, and business continuity in the uncertainty of any applicable infrastructure failure. The addition of the lower levels communication channels is even more effective. reliability and allows a smooth transition of GSM to LoRa. mesh and Ethernet network. As a result, important alerts, commands, and incident information are delivered reliably even in hostile or degraded network situations, ensuring constant presence of situation awareness to both personnel and emergency responders.

In addition, Stealth-AI provides for long-term system evolution with adaptive learning and modular integration. Incident log, detection results, and response patterns can be used to update AI models to perform threat classification better over time. The architecture enables the easy integration of other sensing modalities such as thermal imaging, acoustic sensing or biometric systems that could be used for well-rounded multi-modal threat assessment. Using a combination of intelligent automation, resilient infrastructure and adaptive learning, Stealth-AI lays down a scalable and future-ready system for the next generation of institutional emergency management systems.

II. LITERATURE SURVEY

Due to the increasing threats in schools, governmental institutions, and critical infrastructure, effective Modern emergency response systems have increased the use of intelligent automation. Traditional surveillance platforms are often heavily reliant on manual monitoring, which can result in delayed responses, human error and system failure under high risk circumstances, such as an intrusion or coordinated attempt to jam. As institutions are faced with more sophisticated physical and cyber threats, researchers have highlighted the importance of resilient systems with the ability to detect autonomously, communicate without interruption and provide robust operational continuity. Recent literature emphasizes that resilience of security systems based on the IoT is impacted by the reliability of communication, adversarial resistance and decentralized processing architecture [1],[6],[7]. The findings indicate that sophisticated solutions are essential due to their functionality even at the event of purposely creating a disruptive situation like disruption of power connectivity or jamming of wireless connections.

The additional research points of focus are jamming detection and communication layer resiliency that is paramount in creating operations that are conducted effectively in case of emergencies.. Studies by Wang et al. and Zahra et al. show that IoT environments are extremely susceptible to burst jamming and interference, making them less responsive when reliability is required the most [2],[3].

Further developments with deep reinforcement learning and recurrent neural networks promise to be able to defend against jamming attacks and maintain communication channels with limited conditions [4],[10].

These contributions point to the need for multi-layer connectivity to be used where LoRa mesh networks, software-defined radio (SDR) defenses, and redundant communication paths are used to guarantee that emergency alerts and system commands will be delivered without interruption. Research into Edge AI technologies and LoRa mesh technologies further enables decentralised, low-latency responses to large-scale environments, which play an important role towards operational resilience in crisis situations [9].

Along alongside communication resilience, real time threat detection has become a core aspect of security research. Edge-based learning and artificial intelligence-based surveillance systems are now being adopted increasingly to reduce dependence on cloud and minimize the detection time. The work of Khakurel and Rawat on online learning for physical threat identification can be used to understand how edge AI provides increased situational awareness by enabling quick intruder recognition at a device level [8].

Similarly, the importance of robust, user-centered, notification systems as part of inclusive emergency alert systems and cyber- physical resilience studies has been emphasized [5],[6].

III. SYSTEM ARCHITECTURE

CCTV Camera Stage: Stealth-AI uses CCTV camera stage as its main source of input whereby the camera captures video feed in real time at several surveillance points in the institution. These high resolution feeds are available for most strategic positions and this provides an adequate coverage of key areas.

Video Improvement & Pre-processing Phase: The pre-processing step is applied to enhance the quality of images with enhanced image filtration, noise minimization and contrast optimization to improve the quality of the video before analyzing the threats. This step standardizes the lighting environment and eliminates the relevant frames and conducts the time-based examination to ensure the best possible clarity and accuracy to the next threat detection systems.

Threat Classification/Assignment of Priority stage: This step applies to the computer vision model of SSD and YOLO to recognize and pinpoint possible danger and dangerous articles with given severity scores.

Internal Occupant Notification Stage: Instead of loud and possibly panic-inducing alarms, such alerts are provided at this stage to occupants using visual and digital instruction systems and are discreet and silent. The system provides real-time directions to given safe zones and coordination of the silent lockdown processes in different parts.

Emergency Protocols Stage: It triggers the physical security features such as door locks, barriers, and access control systems and guarantees a coordinated activity to all of the security infrastructure in the whole institution.

Stage of External Responder Notification: The system transmits detailed threat information, threat location and type of threat to emergency services such as police, fire, medical services etc. Real-time intruder heat maps and movement patterns are made available to first responders, so that they have the situation awareness in place before they get there.

Transmission Control Point: This phase keeps checking the primary GSM network connectivity and transfers to the LoRa Mesh connectivity automatically in case of the primary channel failure.

Incident Logging Stage in real-time: The local recording of all the threats detected, system actions, response times, and communication attempts are done in real-time. The system produces rich audit trails with metadata such as timestamps, locations and threat classifications which can be used later after the incident to analyze the system and provide improvement.

Edge AI Processing Device: Dedicated edge computing hardware like Artificial intelligence (AI) enabled embedded systems or on-premise servers (e.g. GPU/TPU enabled devices) to perform real-time computer vision inference with the employment of YOLO and SSD models without the requirements of cloud connectivity

Post-Restoration data sync Stage: This stage is used to broadcast all the locally incident logs and data to the central security operations center when the network connectivity is restored after the outage.

Automated Self Diagnostic/ Learning Level: The system carries out an automated diagnostic of all edge AI devices and communications modules and a check on the general health of the system constantly.

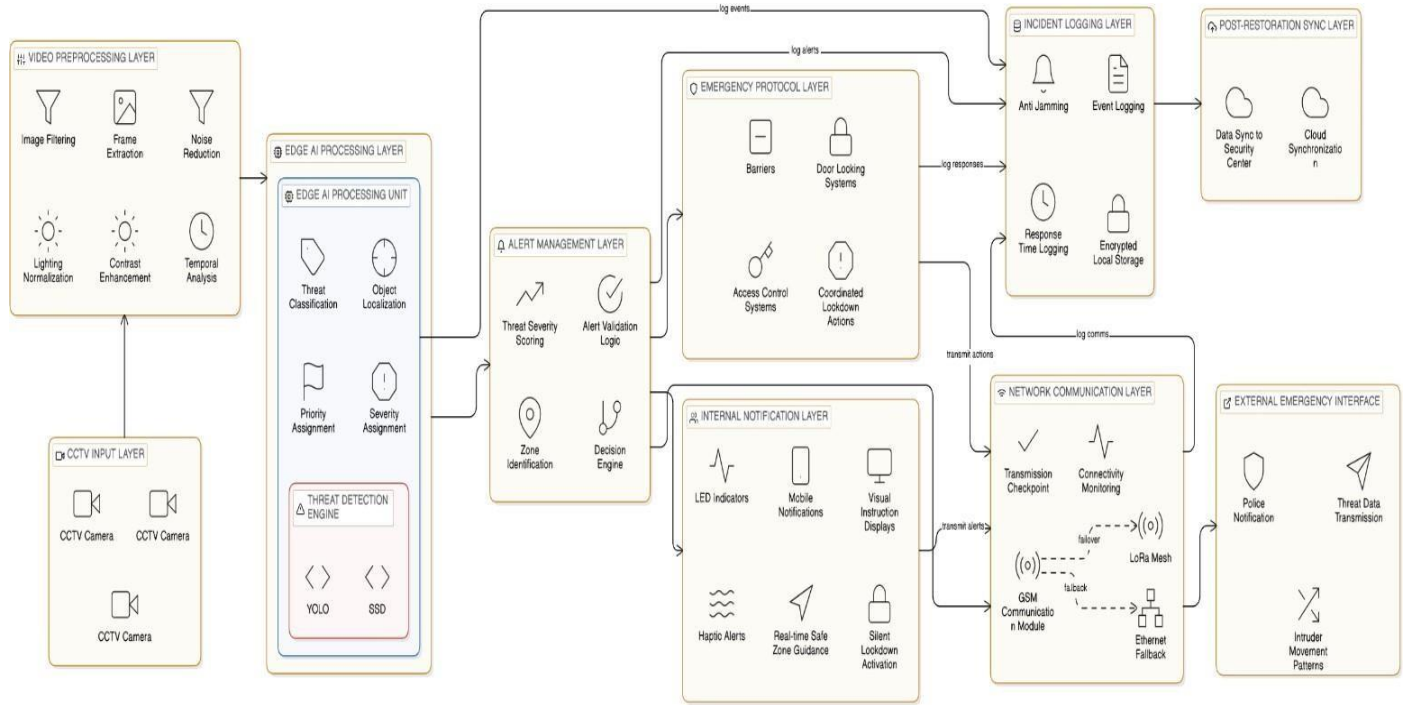


Fig. 1. System Architecture of Stealth AI

IV. REQUIREMENTS AND METHODOLOGY

The successful implementation of Stealth-AI requires a stable CCTV infrastructure, which can deliver quality video in real time, special edge computing hardware, used for the local AI processing, and robust communication, which includes GSM, LoRa mesh, Ethernet fallbacks and SDR-based anti-jamming hardware. Continuous operation is facilitated by backup power sources such as UPS or Generators. Secure local data storage, standardized integration of API with existing security and building management systems and encrypted alert devices are a must! Trained personnel, thorough documentation, and incident logging mechanisms, keep operations running effectively, are compliant and improve systems continually across institutional environments.

A. System Requirements

The hardware requirements that are necessary to ensure effectiveness and reliability of Stealth-AI in all operation situations. Sufficient computing resources like large RAM and multi core processors to handle the intensive computing requirements of real-time video analytics and AI inference that helps in fast threat detection without any delay.

1) Hardware Requirements:

Surveillance nodes or edge servers which has:

- 8GB RAM (16GB or more is recommended for video analytics).
- Multi Core processor, Quad core 64-bit processor (Intel i5 / AMD Ryzen 5 or Equivalent ARM based processor)
- Sufficient storage capacity for local video streaming, model data and system logs (250-1000 gigabytes).

Network Connection:

- Wire ethernet connection for the main communication paths.
- Bluetooth Mesh Modules / Wireless Adapters for Resilient Back Up Networking.
- Stable connection to the internet for updates, remote management, notifications from outside the world if the is available

CCTV Cameras:

- High resolution IP based CCTVs and they have the ability to provide the real time video streaming.
- Cameras should not only be able to work in low-light parameters, but should also be strategically placed to provide good coverage of critical areas of an institution

Communication Modules:

- a. GSM Communication module for alert transmission (primary alert).
- b. Long range, low power back up communication (LoRa Mesh module).

2) Software Requirements:

Web Browser:

- a) Modern web browser (e.g. Google Chrome, Mozilla Firefox, Microsoft Edge, etc.) for accessing the Stealth-AI management console and to monitor the available dashboards that are made available in the cloud.
- b) Operating System: Compatible Operating System for Stealth-AI edge devices and servers OS (Linux, Ubuntu, CentOS, RHEL operating systems and Windows for administrative interfaces and support tools).

AI and Analytics Software:

- a) Pre-installed artificial intelligence frameworks and libraries required for the model inference and training (eg Python, OpenCV, TensorFlow, PyTorch).
- b) Setup of configured computers vision modules for weapons, intruder and anomalies detection (YOLOv8 or similar).

Methodology

methodology for Stealth-AI is modular, resilient and real-world oriented that incorporates AI-based threat detection, secure communication, and continuous system operation.

- 1) **User Interaction Layer:** This is an interface level that consists of a secure dashboard, application and device interface to users including security personnel. They are able to watch the alerts, live feeds, and watch logs live.
- 2) **Signal Interpretation & Alert Management Layer:** This is the logic controller that checks AI detects and deciphers the CCTV data. It determines the type of alert, the zones to which it will be applied, and the route of communication, namely, GSM, LoRa Mesh, or Ethernet. It also makes sure alerts are sent in the correct direction and will change channels in case of failure, so as to continue delivery.
- 3) **Core Detection & execution Layer:** In this, the edge AI is used to scan camera feeds to detect threats in real time. After verification, the system sends alerts, logs activities and contacts other nodes. It is a high-performance, high-accuracy and high-stability layer even in low-connectivity or unstable environments.
- 4) **System Integration Layer:** The layer is where Stealth-AI will be integrated with building controls and external agencies. It handles automatic lockdown, connects to access systems and logs to central servers. The offline activities are emulated and sent to be utilized at a later time thereby ensuring that an operation is not hampered with in complex configurations.
- 5) **Data Management & Storage Layer:** This layer provides secure storage of videos, alerts, logs, and actions of users. The information is encrypted and stored locally or on the cloud. It complies with privacy regulations and is able to provide quick access to audits or investigations.

- 6) **Network & Communication Layer:** This layer is in charge of the GSM, LoRa Mesh, and Ethernet communication. It picks the most suitable channel and encrypts the data and secures data against interference.
- 7) **System Monitoring & Diagnostics Layer:** This layer gets data on the health of software, hardware, networks and AI models. It conducts automatic tests, checks accuracy and resources. Any anomalies raise red flags so that problems can be resolved within a short period of time.

Working principles of the Model

- i. This system applies a CCTV camera and Edge AI to identify the events and alerts. When the main GSM network is lost it changes to backup LoRa mesh. All event information is also stored locally in case it is required and nothing will be lost in case of a connection failure.
- ii. The central system incorporates the best computer vision models, including YOLO, SSD, implemented and LoRaMesh on edge computing to detect threats quickly, based on the results of current research on physical threat recognition in real-time and online learning systems.

Model Used

A. YOLO

YOLO, which stands for "You Only Look Once," is a real-time object detection that uses neural network to identify and locate objects in images with high speed and accuracy.

- i. Real-time object detection: This is utilized in applications where real-time processing is needed, e.g. autonomous vehicles and robotics [2].
- ii. Image splitting: The more modern versions are capable of image splitting, i.e. Object recognition at pixel by pixel.
- iii. Pose estimation: Pose estimation can also be done by some of the versions, which is a task related to the recognition of the main points of objects.

B. SSD

Real-time Object Detection Model used in Stealth-AI to detect threat from CCTVs in real-time feeds in one neural network pass. It identifies objects at multiple scales simultaneously which makes it useful for detecting hidden weapons and intruders and suspicious behavior of varying sizes within surveillance frames. SSD is also quick to run on edge artificial intelligence (AI) devices, allowing for quick local threat detection without using heavy computational resources, which are necessary for the system's resilience against network outages.

C. LoRaMesh

LoRa Mesh (Long Range Mesh) is a decentralized communication network for use in Stealth-AI to send alerts and system data even in case of failure of primary GSM networks. It allows long-range and low-power wireless communication for edge to alert nodes and establishes a mesh topological architecture where the devices communicate relaying messages over various paths for redundancy. LoRa Mesh is especially useful for critical infrastructure because it is not dependent on internet infrastructure so that even during times of network failure, power outages, or malicious attempts to jam signals, critical alerts can still reach the people who need them most, the responders and occupants.

D. Combined usage

Stealth-AI brings together a number of advanced technologies with the goal of building a resilient emergency response system for critical institutions. The system uses SSD (Single Shot MultiBox Detector), a real-time object detection system that detects objects from the CCTV feeds in a single neural network pass at a variety of scale.

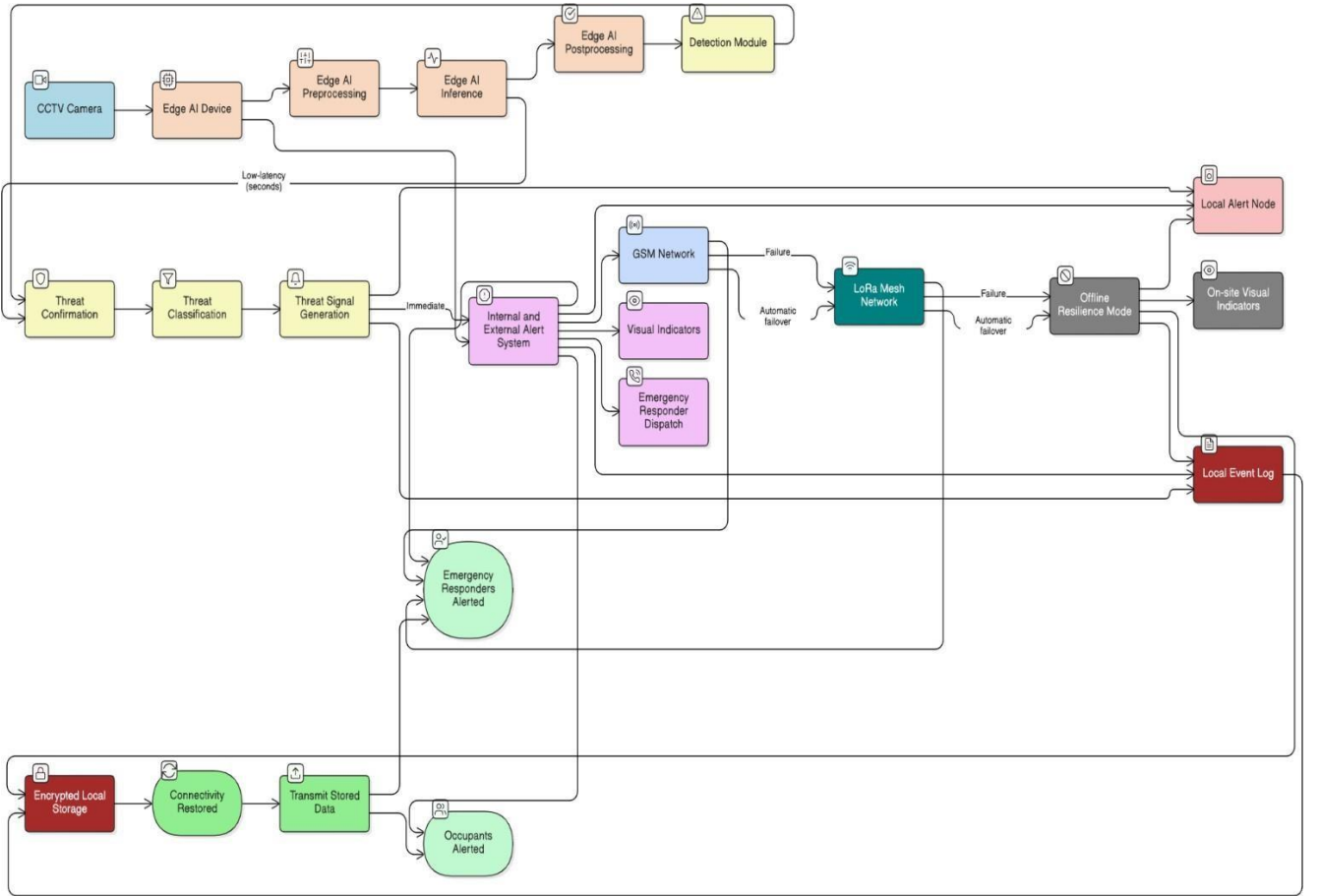


Fig. 2. Workflow Diagram.

Incident Logging Stage in real-time

The local recording of all the threats detected, system actions, response times, and communication attempts are done in real-time. The system generates detailed audit trails with metadata such as timestamps, locations and threat classifications that can be used later after the incident to analyze the system and provide improvement

V. IMPLEMENTATION

The application of the Stealth-AI model started with the constant video surveillance of through strategically located CCTVs. During the testing these cameras worked well and reliably by capturing real-time visual data, even in adverse lighting and crowded conditions.

A. Development Environment

The Development and testing of Stealth-AI system was performed on a Linux-based workstation (Ubuntu 20.04 LTS) and Python as a primary programming language. Visual Studio Code was used as integrated development environment for model development, system integration and testing.

B. Basic Implementation Modules

1) Edge A.I & Threat Detection Module (Back end):

This module is implemented using Python and works on edge AI device to process video streams from CCTV in real-time at the local level. Video frames are captured continuously from the IP-based CCTV cameras and processed pre-hand using OpenCV for noise reduction, frame normalization and contrast enhancement.

2) Alert Management & decision module:

This module takes results of detections from the Edge AI unit and validates possible threats. It assigns priority levels according to the severity of threats, their location and predefined institutional safety policies. The module decides based on the logic of decisions, whether or not to trigger internal alerts, external emergency notification, or both. It also coordinates silent activation of the lockdown, as well as guidance instructions to the occupants. The module oversees the selection of the communication channels and initiates failover mechanisms where needed.

3) Internal Alert Module (Frontend):

This module is in charge of the silent alerts to occupants provided by visual indicators, mobile notifications, and haptic feedback devices. Real-time visual guidance is shown to help occupants to designated safe zones in case of emergency. The interface dynamically changes based on the incoming alert data and changes in the state of the system. The design does not use audible alarms to prevent panic and ensure that controlled evacuation or lockdown procedures are followed.

4) Emergency Responder Interface to External:

This module can send verified threat information to external emergency services such as police, fire and medical services that include the type, location and severity of the threat. Real-time intruder movement information and real-time heat maps are shared and help enhance situational awareness before a responder arrives.

5) Incident Logging and Data Storage Module:

All detected threats, alerts, actions and communication attempts are recorded locally in encrypted storage. Logs stamp and time, camera ID, threat classification and response action for audit and forensic analysis, when there are interruptions to the network.

VI. RESULTS AND DISCUSSION

The implementation of Stealth-AI model began from the continuous video surveillance assisted with strategically placed CCTVs. During the testing, these cameras were quite capable of capturing actual visual data in real-time even in unfavorable light and crowded situations. Image quality and coverage was found to be key in enabling the use of robust AI analysis with higher resolution feeds directly improving detection accuracy. The coupling of CTV as the main source of data made a good background for the next process of threat identification. Edge AI devices played a pivotal role in the processing and analyzing the camera streams, using in it advanced computer vision algorithms meant for fast recognition of weapons and intrusive behavior. Results showed that the on-device inferencing decreased the latency and the over dependence on external servers. The edge devices provide a consistent detection on time which resulted in a high confidence and low false alarm rates. Performance remained stable even under changing network conditions verifying the reliability and effectiveness of decentralized architecture of AI. Upon threat detection the internal and external alert modules of the system were coordinated in their response.

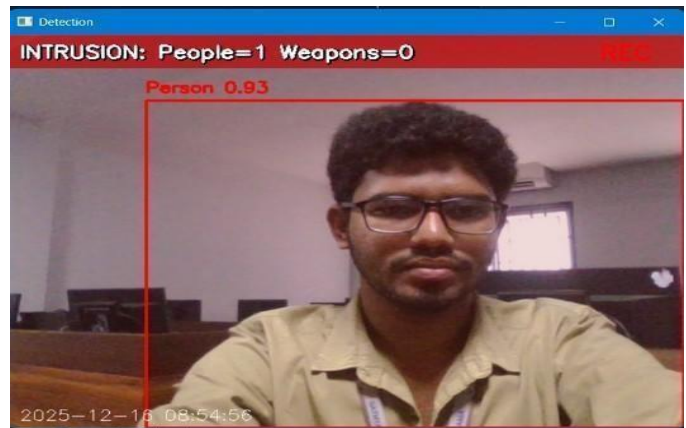


Fig 6.1 Detection Output

VII. CONCLUSION AND FUTURE SCOPE

The Stealth-AI emergency response system project is a successful demonstration of a transformational approach to institutional safety and crisis management based on modern AI, resilient communication technologies. From the first threat detection to the last alert communication the solution is a complete solution combining a smart edge AI-analysis, discrete internal and external notification mechanisms and a strong multi-modal communication protocol. The layered design offers uninterrupted operation even in event of power outage, signal jamming and network disruption conditions, which are the usual shortcomings found in the conventional surveillance and alert systems command.

Future Scope

Future improvements of Stealth-AI are a combination with sophisticated biometric identification systems for better tracking of intruders and multi-institutional coordination by so-called federated learning networks. The system can be extended to take care of the predictive threat modeling based on historic data of incidents in order to take proactive security and not purely reactive

security measures. Implementation in a variety of institutions such as hospitals, government buildings and public transportation systems is a great potential for scalability with custom threat classification models for different types of facilities. Future developments should include fusion of multimodal sensors to add thermal imaging, threat detection by sound and behavioural analysis for complete threat identification.

REFERENCES

- [1] Bures, Miroslav, et al. "Factors impacting resilience of internet of things systems in critical infrastructure." 2022 IEEE World AI IoT Congress (AIIoT). IEEE, 2022.
- [2] Wang, Shao-Di, Hui-Ming Wang, and Peng Liu. "Fast detection of burst jamming for delay-sensitive Internet-of-Things applications." *IEEE Transactions on Wireless Communications* 22.6 (2022): 3955-3967.
- [3] Zahra, Fatima Tu, Yavuz Selim Bostanci, and Mujdat Soyuturk. "Real- time jamming detection in wireless IoT networks." *IEEE Access* 11 (2023): 70425-70442.
- [4] Ali, Abubakar S., Shima Naser, and Sami Muhaidat. "Defeating proactive jammers using deep reinforcement learning for resource- constrained IoT networks." 2023 IEEE 34th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC).
- [5] Kshirsagar, Utkarsha, et al. "Inclusive Emergency Alerts: A Cloud-Based Haptic and Visual Notification System for the Deaf and Hard of Hearing." *Proceedings of the International ISCRAM Conference*. 2025.
- [6] Aghazadeh Ardebili, Ali, et al. "IoT-Driven Resilience Monitoring: Case Study of a Cyber-Physical System." *Applied Sciences* 15.4 (2025): 2092.
- [7] Dede, Georgia, et al. "Cybersecurity as a contributor toward resilient internet of things (IoT) infrastructure and sustainable economic growth." *Information* 15.12 (2024): 798.
- [8] Khakurel, Utsab, and Danda B. Rawat. "Real-time physical threat detection on edge data using online learning." *IEEE Consumer Electronics Magazine* 13.1 (2023): 72-78.
- [9] Vermesan, Ovidiu, et al. "Edge AI LoRa Mesh Technologies." *Advancing Edge Artificial Intelligence*. River Publishers, 2024. 1-42.
- [10] Ali, Georges Kaddoum, and Walid Saad. "Recurrent-neural-network- based anti-jamming framework for defense against multiple jamming policies." *IEEE Internet of Things Journal* 10.10 (2023): 8799-8811